



3D Coupled Modelling of a Series-Connected Bifilar Cu-Stabilised REBCO Resistive SFCL

H-formulation, solid heat transfer and external circuit solved together on a 50 mm representative segment of four tape runs, 400 V source model

1 Introduction

Problem. Distributed generation raises the fault level at a 400 V feeder built for a lower one.

Answer. A resistive SFCL in series with the feeder (Fig. 1(a)–(c)) is invisible in normal operation and develops resistance within the first cycle [1]–[4].

Difficulty. Current, resistance, temperature and cooling set one another within one cycle, so they must be solved together.

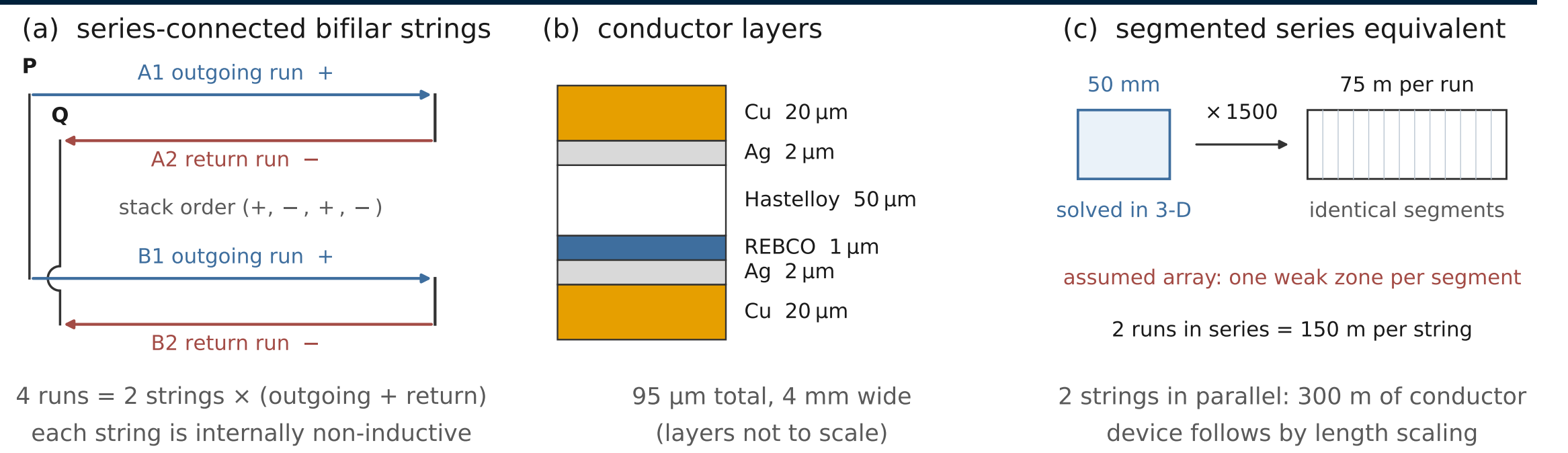


Fig. 1(a) Two bifilar strings in parallel, each non-inductive. (b) Layers. (c) Segment array.

2 Methodology

Electromagnetic. H-formulation [6] with the power law $E = E_c (|J|/J_c)^n$ [5], anisotropic Kim-type $J_c(T, B)$ and a localised weak zone.

Thermal. Conduction across all conductor layers from the Joule source $Q = \mathbf{E} \cdot \mathbf{J}$, properties temperature dependent.

Cooling. Boiling-curve boundary $q = h_{LN_2}(\Delta T) \Delta T$ [7] into a 77 K bath.

Circuit and mesh. 400 V Thévenin source, $X/R = 3.79$, fault and clearing smoothed, clearing time swept (Fig. 1(d)); a 50 mm representative segment of four tape runs on a 5 mm pitch, 119,184 elements, the REBCO film its own 1 μm layer.

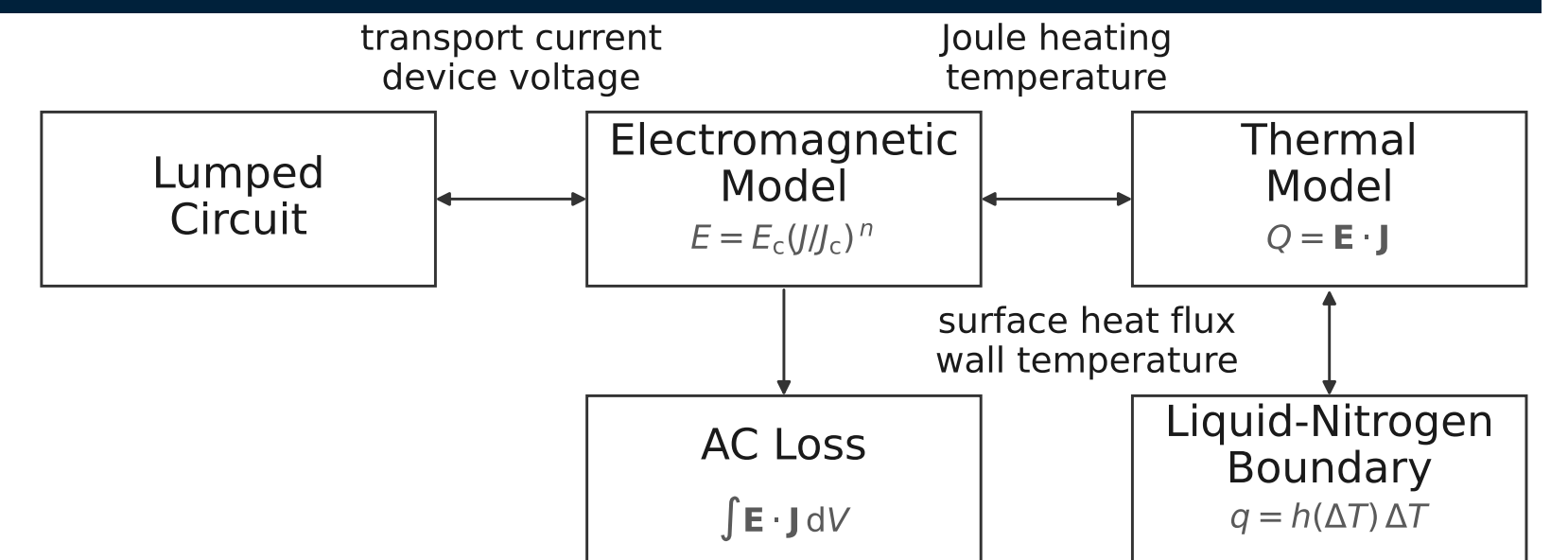


Fig. 1(d) The three sub-models and the boiling boundary as one system.

3 Results & Analysis

Limiting. The first fault peak falls from a prospective 2931 A to 456.1 A, a reduction of 84.4 %.

Temperature. The volume average peaks at 80.48 K and the layer maximum at 81.89 K, 8.1 K under the 90 K critical temperature.

Clearing. Over settings of 20 to 80 ms, measured from fault inception to current interruption, the hottest sampled point falls below 78 K within 15.2 to 25.0 ms of interruption.

Winding and loss. Counter-winding moves the field rather than removing it, cutting stored magnetic energy about 2.8 times; transport a.c. loss at rated current is 5.27 W over the derived 300 m of tape.

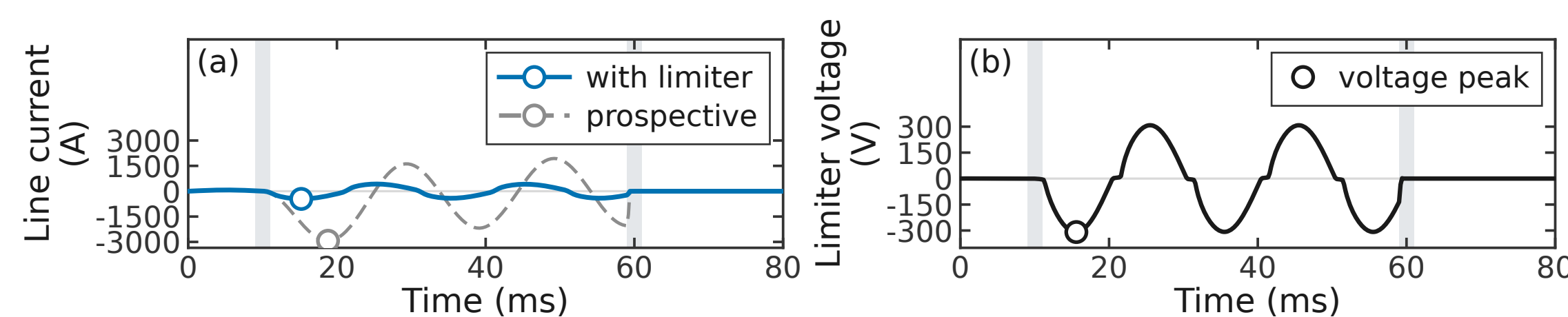


Fig. 2(a) Line current with and without the limiter; 84.4 % is the largest of four resolvable half cycles (73.7 to 84.4 %), at the 10 ms inception phase computed. (b) Limiter voltage, peak 308.9 V, 0.400 ms after the current peak.

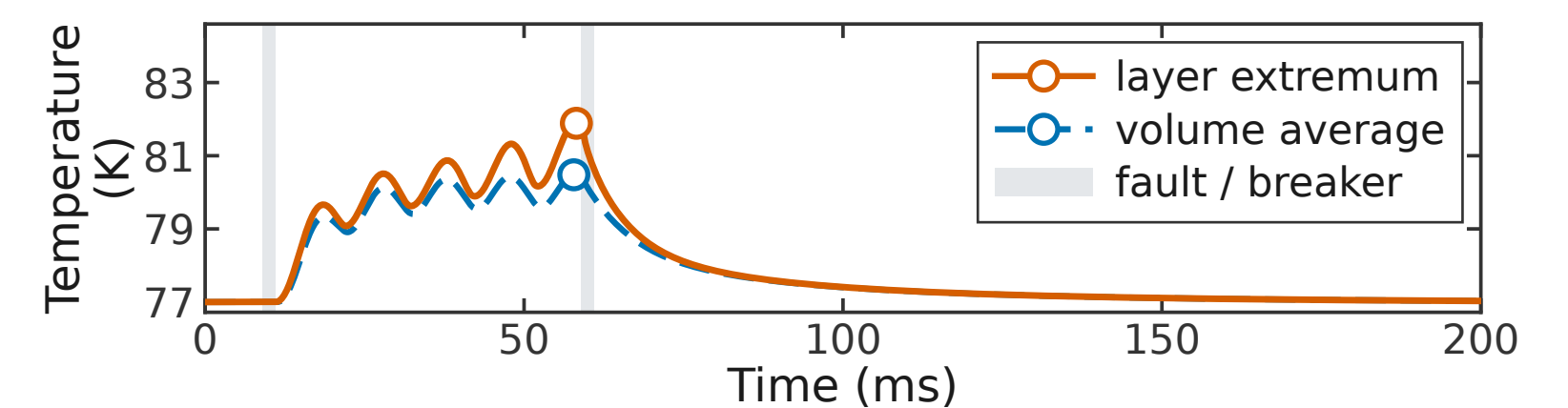


Fig. 2(c) Temperature, double reported; the layer trace is a maximum operator over a sampled selection.

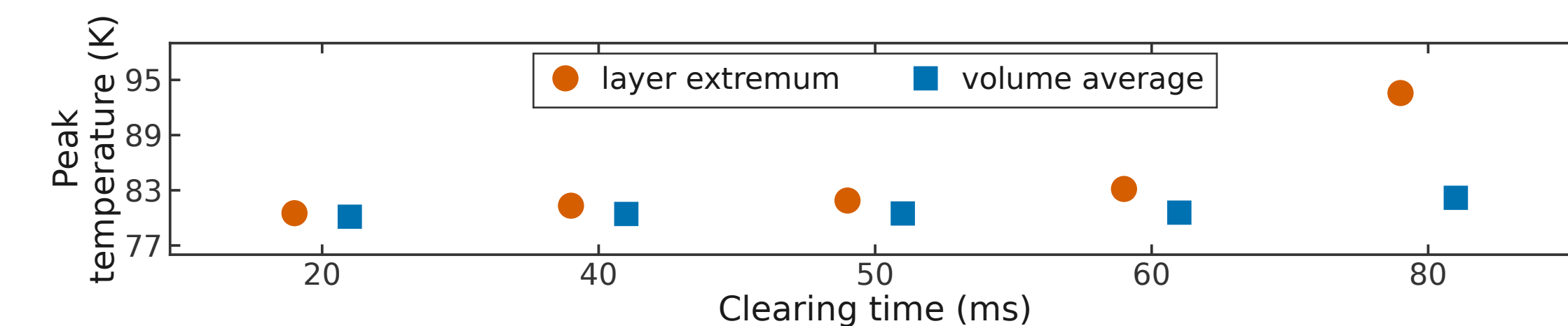


Fig. 2(d) Peak temperature against breaker delay, one run per setting and uninterpolated; layer extrema 80.5 to 93.6 K, volume averages 80.1 to 82.2 K. Settings past 80 ms are unresolved.

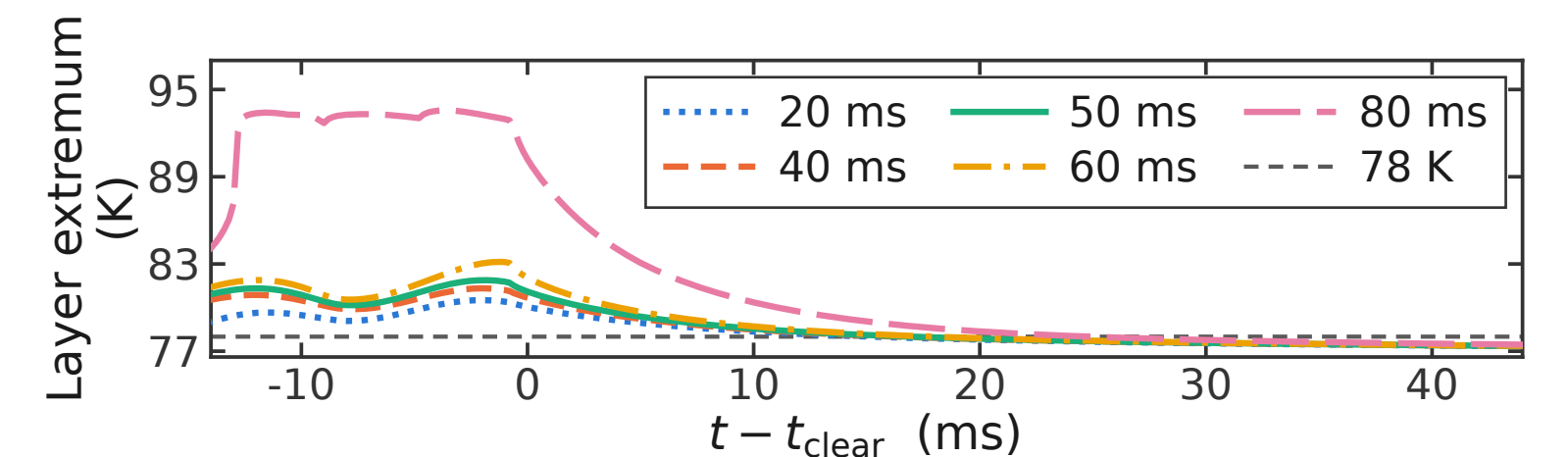


Fig. 2(e) Cooling after interruption, referred to each clearing instant.

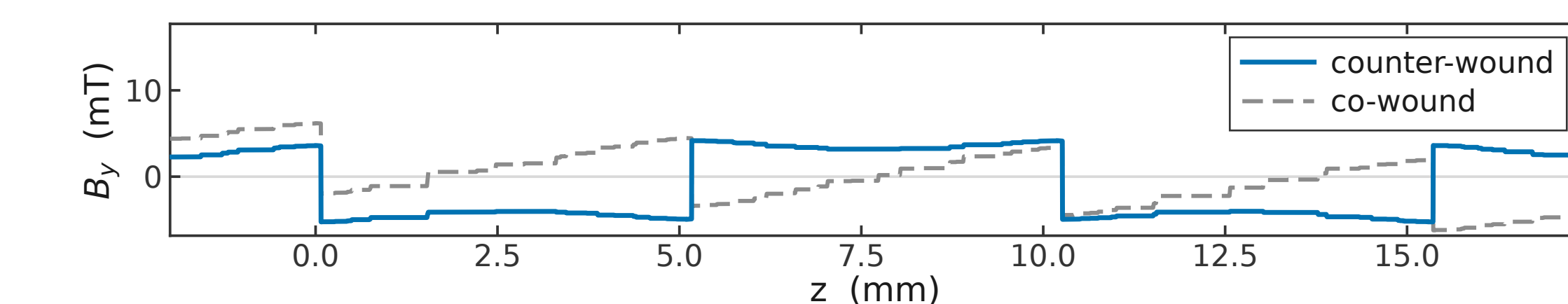


Fig. 2(f) B_y through the four-tape stack at rated current, both winding senses. The far-field share of $\int B^2 dV$ falls from 82 % to 33 % as inter-tape energy rises 1.3 times; 2.8 is a lower bound growing with the air domain.

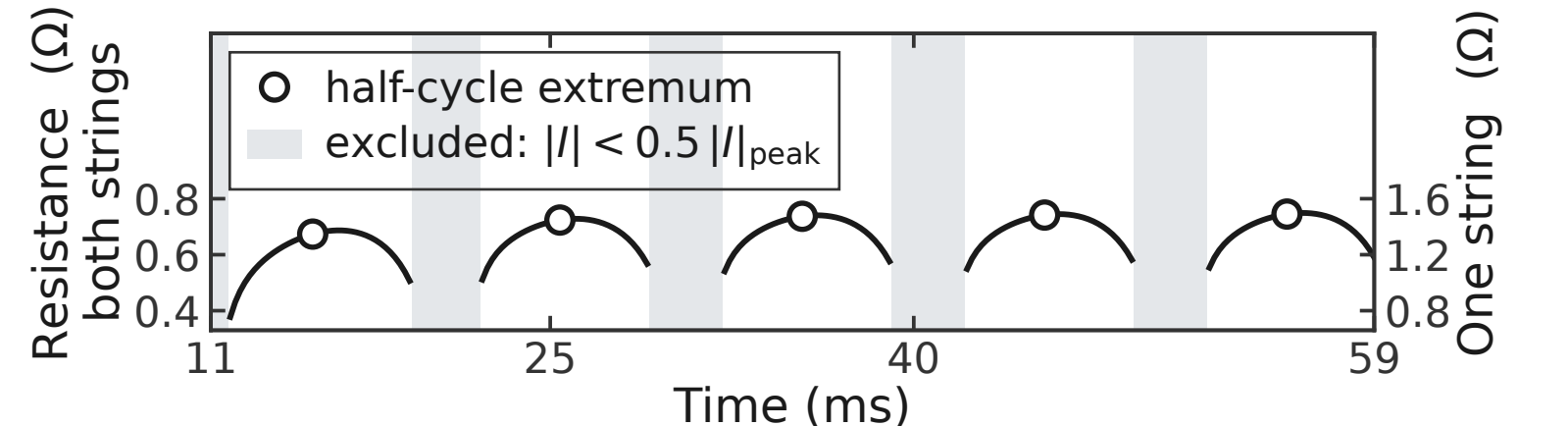


Fig. 2(g) Device resistance, both strings in parallel: 0.67 to 0.75 Ω where $|I| \geq 0.5 |I|_{peak}$, the terminal ratio being inductance-dominated near current zeros.

4 Conclusion

A fully coupled 3-D model of a series-connected bifilar REBCO limiter on a 400 V feeder cuts the first fault peak by 84.4 %, holds the conductor 8 K below its critical temperature, and brings the hottest sampled point back below 78 K within 25 ms of every interruption computed. The 3.5 K volume-average and 4.9 K local rises are both reported because they govern post-clearance cooling and local quench propagation respectively. Counter-winding buys a 2.8-fold drop in stored magnetic energy at the cost of a stronger field between the tapes.

Currently, all verification is confined to the internal model, as experimental data is not yet available. Future work will focus on experimental validation, beginning with the confined-gap boiling boundary, to bridge the gap between design and practical application.

References

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